EARLY-PHASE CLINICAL PHARMACOLOGY · BIOSTATISTICS DELIVERABLES PIPELINE · 132 DELIVERABLES · REFERENCE MAP

Protocol review → CSR draft

S1 Study Design & Protocol

- Protocol biostat input
- Estimands → endpoints
- Sample-size justification
- Randomization design
- DSMB / SRC decision rules

GATE **Protocol approved**

S2 Planning & Specifications

- Statistical Analysis Plan
- TLF shells / mock-ups
- ADaM & SDTM specs
- Randomization schedule
- Data-review & QC plans

GATE **SAP & specs final**

S3 Data & CDISC

- SDTM datasets + aCRF
- ADaM datasets
- Define-XML + cSDRG/ADRG
- Conformance (Pinnacle 21)
- Database lock + audit trail

GATE **Database lock**

S4 Analysis & QC

- PK conc & NCA TLFs (Phoenix)
- Safety / PD TLFs
- popPK / exposure-response
- Independent double-programming
- Unblinding (controlled)

GATE **Outputs QC-passed**

S5 Reporting & CSR

- CSR methods & results text
- In-text tables / figures
- CSR appendices (16.x)
- Define + guides → eCTD m5
- Cross-doc consistency QC

GATE **CSR draft delivered**

CDISC data-standards track: SDTM → ADaM → Define-XML + cSDRG / ADRG + conformance (runs through Stages 3–5)

Quality track: independent double programming · engine-of-record register (software and version) · audit trail (21 CFR Part 11) · named sign-off (all stages)

Double programming every submission-grade output: an independent second program and a logged reconciliation

Validated engines reported numbers from validated SAS/R programs; NCA from Phoenix WinNonlin or PKNCA

Named sign-off a qualified statistician reviews and signs each deliverable

Reference map, not a track record: how each deliverable would be produced and checked. A proposal for AI-assisted drafting under controls is an appendix to the reference document.

132 deliverables across 5 stages · full register with every field = the .xlsx · stage-by-stage detail = the Word/PDF reference · standards as of October 2026 (CDISC SDTMIG v3.4 / ADaMIG v1.3 / Define-XML v2.1; ICH E3·E6(R3)·E9(R1)·M11; 21 CFR Part 11).